VisionGuard — Easy Technical Guide

Simple Step-by-Step Explanation of How VisionGuard AI Works

■ What is VisionGuard in Simple Words?

VisionGuard is like a **smart AI security guard inside a camera**. It automatically watches live video, finds people, and checks if anyone is smoking in no-smoking areas. When it catches someone smoking, it automatically takes photos, saves evidence, and alerts the dashboard instantly.

1. Simple Technology Stack (What Powers VisionGuard?)

Component	What We Used	Simple Explanation
AI Hardware	NVIDIA RTX 4050 GPU	Fast graphics card used to run AI models at 50+ frames per second.
Person AI Model	YOLO11n (Nano)	Small AI model that detects human bodies and gives each person a ID number.
Cigarette AI Model	Custom YOLO26s (STAL)	Trained for **60 to 150 rounds (epochs)** on GPU to spot small cigarettes.
Backend Server	FastAPI & Python	Main server program that handles camera streams, alerts, and settings.
Database	SQLite (visionguard.db)	Database file that stores violation photo evidence, dates, and times.
Web Dashboard	HTML & JavaScript	Live dashboard website where users view live feed, alerts, and photos.

2. How the 2 AI Models Work Together

VisionGuard uses **two AI brains** working at the exact same time:

Brain #1 — Person Finder (YOLO11n):

- Scans the camera feed to find human bodies.
- Draws a **green box** around each person and tracks them with a number (e.g. Person #1).

Brain #2 — Cigarette Finder (Custom YOLO26s):

- Trained on an NVIDIA RTX 4050 GPU for **60 to 150 epochs (training cycles)**.
- Specifically trained to detect small, thin cigarettes, lit tips, and smoke.
- Draws an **orange box** around cigarettes when found.

3. How VisionGuard Prevents False Alarms (Spectacles, Pens & Folds)

Standard AI models sometimes mistake spectacle frames, glasses stems, pens, or neck shadows for cigarettes. VisionGuard uses **4 Smart Rules** to stop false alarms:

False Alarm Cause	Smart Rule Solution	Why It Works Simply

Spectacle frames, glasses arms, eyes, ears	**No Glasses/Eyes Rule** (Top 47% head block)	Blocks the upper 47% of the head. A cigarette is never placed on someone's eyes or glasses stem!
Neck shadow folds & shirt collar lines	**No Neck Shadow Rule** (Lower neck block)	Rejects square shadow folds under the mouth.
Pens, cables, lollipop sticks	**Shape & Confidence Rule** (Conf >= 50%)	Rejects weak guesses and thin non-cigarette shapes.
Cigarette picture on a poster or chair	**Person Overlap Rule** (Containment >= 10%)	Cigarette MUST belong to a real person standing in view.

4. Step-by-Step: What Happens When Someone Smokes?

Here is what happens inside the system in less than 1 second when someone smokes:

1. Camera Capture

The camera records live video continuously and sends frames to the server.

2. AI Scanning

The 2 AI models scan the frame: one finds the Person, the other finds the Cigarette.

3. Smart Rule Verification

The system checks: Is the cigarette at the mouth? Is confidence over 50%? Is it NOT glasses?

4. Multi-Frame Double Check

The AI waits for **3 consecutive frames** of smoking to ensure it is not a temporary glitch.

5. Automatic Photo Saving

Once confirmed, it automatically takes **4 to 6 evidence photos** with boxes drawn around the smoker.

6. Database Archiving

Saves the event record, time, date, and photo folder path into SQLite database (visionguard.db).

7. Live Alert on Dashboard

The web dashboard instantly displays red alert cards, updates counters, and shows the evidence photos.

5. Dashboard Features

- **Live Video Feed:** Watch live video with green boxes for people and red boxes for active smoking.
- **KPI Counters:** Real-time counters showing total People, Smokers Now, FPS, and Confirmed Count.
- **Violation Photos Tab:** View multi-frame evidence photo sets taken during violations with zoom view.
- **Summary Analytics:** Color-coded progress bars showing confirmed vs rejected violations.
- **Settings Controls:** Easily adjust AI sensitivity sliders live without restarting.

Status: *VisionGuard is fully operational, GPU-accelerated on NVIDIA RTX 4050, and pushed to GitHub (kanth071/Smoking-detection).*